

Recursion

1) Find Out the Applications of Recursion.

A Recursion is a programming technique where function calls itself.

1) Factorial Calculation

2) Fibonacci Series

3) Tree Traversal

4) Graph Traversal

5) Searching and Sorting

6) Backtracking Problems

→ Solving Sudoku

→ N-Queens Problem

→ Maze Solving

→ Knight's Tour

7) Tower of Hanoi

8) Mathematical Computations

→ Greatest Common Divisor

→ Power Calculation

→ Exponentiation

9) Directory Traversal

→ Exploring folders and sub folders recursively

10) String Processing

Real-life applications

- 1) File Explorer
- 2) Family Tree
- 3) Website Navigation
→ Web crawlers [like search engine] recursively visit webpages and follow links to other pages.
- 4) Google Maps
→ Maps use recursive algorithm in route planning.
- 5) Decision Trees in AI.
- 6) Menus in Apps and Websites.

2) Qs Recursion still used in the present programming era. If so where?

A Yes, Recursion is still widely used in modern programming.

1. File system
2. Web Browsers and Search Engines
3. Data Structures
4. Algorithms
5. Artificial Intelligence
6. Compilers and Programming
7. Backtracking Problems
8. Graphics and Games

9. Databases

10. Mobile and Web Applications

3) What are the languages that support recursive calls?

A 1) C

2) C++

3) Java - Recursion used for trees, graphs

4) Python - Supports recursion

5) JavaScript - Web development and algorithms

6) C# - Supports recursion in .NET applications

7) Go (Golang)

8) Swift

9) Kotlin

10) Rust

11) PHP

12) Ruby

13) R

14) Haskell

15) Lisp/Scheme